ARITHMETIC AND GEOMETRY OF ALGEBRAIC CYCLES (PHD STUDENT SEMINAR)

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1. Introduction

In topology we define homology groups. Intuitively, we "linearize" a topological space by looking at formal sums of certain geometric building blocks (in singular homology, n-cycles, which are certain n-simplices) and we look at them up to an equivalence relation based on boundaries, that is, some connection on a higher dimension.

For example, the homology of the complex projective space $\mathbb{P}^n_{\mathbb{C}}$ is known to be

$$\mathrm{H}_k(\mathbb{P}^n_{\mathbb{C}}) = \begin{cases} \mathbb{Z} & k \leq 2n, k \equiv 0 \mod 2 \\ 0 & \text{otherwise} \end{cases}$$

Specifically, it is known that $\mathrm{H}_{2k}(\mathbb{P}^n_{\mathbb{C}})$ is generated by $[V_k]$ - the fundamental class of a linear subspace of complex dimension k.

So in this case, the homology classes have representatives that are algebro-geometric in nature. Unfortunately, the equivalence relation is still not very algebraic. Let's try recasting everything in algebraic geometry terms, but first:

Definition 1 (lying a bit)**.**

- An *algebraic variety* over $\mathbb{C}$ is a subset of $\mathbb{P}^n_{\mathbb{C}}$ cut out by a system of polynomial equations.
- A *morphism of algebraic varieties* is a map defined by polynomial functions.
- the dimension of an algebraic variety is it's dimension as a complex manifold (at least locally where it's non-singular)

Now with these definitions, we can suggest another way of interpreting homology:

Definition 2 (Algebraic and Rational Equivalences (lying a bit))**.** Given an algebraic variety X, we say that two k-dimensional closed subvarieties Y_1, Y_2 of X are *algebraically equivalent* if there is some $(k+1)$-dimensional $W \subseteq X$, a curve C, and a map $\phi\colon W \to C$ such that $\phi^{-1}(p_1) = Y_1$ and $\phi^{-1}(p_2) = Y_2$ for some $p_1, p_2 \in C$.

If we allow only $C = \mathbb{P}^1$, we get the coarser relation *rational equivalence.*

Question. Given $Y_1, Y_2 \subseteq \mathbb{P}^n_{\mathbb{C}}$, is it true that $[Y_1] = [Y_2]$ if and only if $Y_1 \sim_{rat} Y_2$?

Answer: Yes! one direction is the following claim which is true in general

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Claim 3. *([Ful84] Proposition 19.1.1) Rationally equivalent implies homologous.*

Let's sketch an answer for the other direction. Assume $[Y_1] = [Y_2]$, in particular, this means that Y_1, Y_2 have the same degree (the degree of the polynomials defining them).

Starting with two lines (that is, degree 1) in $\mathbb{P}^2_{\mathbb{C}}$ (they are homologous because we computed the homology of projective space): If $\ell_1, \ell_2 \subseteq \mathbb{P}^2_{\mathbb{C}}$ are two distinct lines, then they intersect in a single point P. Choose some arbitrary line ℓ_3 not passing through P and define $\phi\colon \mathbb{P}^2_{\mathbb{C}} \to \ell_3 \cong \mathbb{P}^1$ the projection from the point P onto ℓ_3 (draw a picture! mention that we don't really need ℓ_3 but actually the Grassmannian) is the map we are looking for.

For curves of general degree, assume Y_i is defined by a polynomial f_i. the function we are looking for is $[f_1 \colon f_2]$. the fiber over $[0:1]$ is Y_1 and the fiber over $[1:0]$ is Y_2.

So for the homology of $\mathbb{P}^n_{\mathbb{C}}$, we have successfully recast everything in terms of algebraic geometry. This suggests a recipe we should follow in general

Definition 4. Let X be an algebraic variety.

- Denote by $Z_k(X)$ the set of formal sums of closed subvarieties of dimension k (called algebraic cycles).
- Denote by $\mathrm{CH}_k(X) \coloneqq Z_k(X)/\sim_{rat}$ - the k-th Chow group of X. Strictly speaking, we can't use the definition of rational equivalence as stated above, but need to extend it linearly, transitively, and make it account for multiplicities (i.e if a function ϕ has a single zero P but with multiplicity 2, then $\phi^{-1}(0) = 2P$ in $Z_0(X)$.
- If X is smooth, we have the *cycle class map* $\gamma\colon \mathrm{CH}_k(X) \to H_{2k}(X)$ sending a closed subvariety to its fundamental class (and extending linearly).
- We have a degree map $\deg\colon Z_k(X) \to \mathbb{Z}$ defined by $\sum_{n=1}^N a_n P_n \mapsto \sum_{n=1}^N a_n$. it induces a well-defined $\deg\colon \mathrm{CH}_k(X) \to \mathbb{Z}$. We denote its kernel by $\mathrm{CH}_k(X)_0$ - the degree 0 cycle classes.

Remark. Studying this cycle class map relates to Hodge theory and in particular to the famous Hodge conjecture.

Remark. The Chow group should be closely related to homology. In different contexts we expect to have cycle class maps into appropriate (co)homology theories. For example, in a p-adic context, there is a map into Étale cohomology with p-adic coefficients. In this sense, one says that the Chow group contains the geometric information of the algebraic variety. In turn this means that the Chow group is a suitable framework for the construction of *motives*, whatever that is.

2. 0-th Chow group of a curve

Let's investigate $\mathrm{CH}_0(C)$ where C is a curve, the important invariant is the genus of the curve g. The degree map induces a splitting $\mathrm{CH}_0(C) = \mathbb{Z} \times \mathrm{CH}_0(C)_0$, so the interesting part is investigating degree 0 cycle classes.

If $g = 0$ then $C = \mathbb{P}^1_{\mathbb{C}}$ that we already dealt with.

If $g \geq 1$ We show that $\mathrm{CH}_0(C) \cong \mathbb{Z} \times \mathbb{C}^g/\Lambda$ where $\Lambda \subseteq \mathbb{C}^g$ is a lattice.

- $\mathrm{H}^0(C, \Omega^1_C)$, the (linear) space of global holomorphic 1-forms is g dimensional. Pick $\omega_1, \ldots, \omega_g$ a basis.
- $\mathrm{H}_1(C, \mathbb{Z}') \cong \mathbb{Z}^{2g}$. Pick a generating set $\gamma_1, \ldots, \gamma_{2g}$.
- for every $1 \leq i \leq 2g$, define a vector $\Gamma_i \in \mathbb{C}^g$ by integrating the various forms we picked on γ_i:

$$\Gamma_i \coloneqq \begin{pmatrix} \int_{\gamma_i} \omega_1 \\ \vdots \\ \int_{\gamma_i} \omega_g \end{pmatrix}$$

- $\Lambda \coloneqq \langle \Gamma_1, \ldots, \Gamma_{2g} \rangle$ is the lattice we are looking for. Denote $\mathrm{Jac}(C) \coloneqq \mathbb{C}^g/\Lambda$.
- Given $P_0 \in C$ define $\iota_{P_0} \colon C \to \mathrm{Jac}(C)$ (called the Abel-Jacobi map) by

$$\iota_{P_0}(P) = \begin{pmatrix} \int_{P_0}^{P} \omega_1 \\ \vdots \\ \int_{P_0}^{P} \omega_g \end{pmatrix}$$

  Note that the map is well-defined (does not depend on the choice of path from P_0 to P) exactly because we modded out by Λ.

The following theorem seals the deal

Theorem 5 (Abel-Jacobi)**.** *Pick a point $P_0 \in C$. the map $\iota_{P_0} \colon C \to \mathrm{Jac}(C)$ extends linearly to an isomorphism $\mathrm{CH}_0(C)_0 \xrightarrow{\sim} \mathrm{Jac}(C)$.*

The punch line: $\mathrm{CH}_0(C)_0$ is an abelian variety of dimension g (that is, an algebraic variety with a group structure, like a Lie group), and $\mathrm{CH}_0(C) \cong \mathbb{Z} \times \mathrm{Jac}(C)$. Furthermore, fixing some $P_0 \in C$, the map ι_{P_0} embeds C into its Jacobian. It turns out that it's very useful that you can canonically embed a curve inside an Abelian variety.

3. How large are Chow groups? Mumford's theorem

Question. We "saw" that for $X = \mathbb{P}^n_{\mathbb{C}}$ the cycle class map is an isomorphism. Then we saw that for curves of positive genus, the chow group is much bigger than the homology group. What is true in general?

The phenomenon we saw before, where $\mathrm{CH}_0(C)_0$ is an Abelian variety was formalized by Francesco Severi, an Italian mathematician working in early 1900's. Severi "proved" that the Chow group of a general algebraic variety is finite dimensional in the following sense:

Definition 6. We say that $\mathrm{CH}_0(X)$ is finite-dimensional if $\exists n \in \mathbb{N}$ such that the map $\mathrm{Sym}^n X \times \mathrm{Sym}^n X \to \mathrm{CH}_0(X)_0$ defined by $(A, B) \mapsto A - B$ is surjective.

Without going into details, this means exactly that $\mathrm{CH}_0(X)_0$ is an abelian variety, as we saw in the case of curves.

Unfortunately, this theorem was completely false. Mumford proved the following

Theorem 7 ([Mum69])**.** *Let F be an algebraic surface over $\mathbb{C}$ with geometric genus > 0 (dimension of space of global differential 2-forms), then $\mathrm{CH}_0(F)$ is NOT finite-dimensional.*

The Chow groups are large exactly in this sense.

4. Ceresa Cycle

Given a curve C, and a point P, we can embed C into it's Jacobian $\iota_P : C \to J$ by $\iota_P(Q) = [Q] - [P]$. We then get another embedding by $-\iota_P$.

The simplest and most natural cycle we can think of is probably the following.

Definition 8. The Ceresa cycle of C is defined by

$$\kappa_P(C) \coloneqq [\iota_P(C)] - [-\iota_P(C)] \in \mathrm{CH}_1(J)$$

it could be shown that for $\gamma\colon\ \mathrm{CH}_1(J) \to \mathrm{H}_2(J)$ the cycle class map, $\gamma(\kappa_P(C)) = 0$, that is $\kappa_P(C) \in \mathrm{CH}_1(J)_{hom}$ the subgroup of homologically trivial cycle classes.

The Ceresa cycle actually has very deep importance in various directions, but it's also interesting simply because it is relatively easy to research in a group that is largely mysterious.

Theorem 9 (Ceresa, [Cer83])**.** *The Ceresa cycle of a "general" curve with genus ≥ 3 has infinite order in $\mathrm{CH}_1(J)_{hom}/\sim_{alg}$.*

We start by investigating the Ceresa cycle for the following kind of curves

Definition 10. A curve C with genus $g \geq 2$ is called hyperelliptic if it has a degree 2 map to $\mathbb{P}^1$. It can be shown that C has a a model of the form $y^2 = f(x)$ where f is a polynomial of degree $2g+1$ or $2g+2$. in this model, the degree 2 map is $(x, y) \mapsto (x, y^2)$ onto $y = f(x)$ which is isomorphic to $\mathbb{P}^1$.

Remark. The family of hyperelliptic curves is very large: for $g = 2$ every curve is hyperelliptic. For $g \geq 3$, the general curve is not hyperelliptic, but the locus of hyperelliptic curves of genus g inside the moduli space of genus g curves has fairly high dimension. ($2g - 1$ inside $3g - 3$)

Claim 11. *Ceresa cycle of a hyperelliptic curve is trivial.*

The proof is not hard but has some technical parts we didn't mention. The point is that hyperelliptic curves have a special geometric property and we use the involution on them, so the reason the claim is true is "because of geometry".

Proving vanishing or non-vanishing of the Ceresa cycle has been a matter of interest recently, for example

Theorem 12 (Laga–Shnidman, [LS25])**.** *There is an injection $\mathrm{H}^1(J)(-1) \hookrightarrow \mathrm{H}^3(J)$ given by cupping with the principal polarization. Denote by $H^3(J)_{\mathrm{prim}} \coloneqq H^3(J)/H^1(J)(-1)$ its cokernel. If $H^3(J)_{\mathrm{prim}}^{\mathrm{Aut}(C)} = 0$, then $\kappa_P(C)$ is torsion.*

and on the other hand there is work to certify when the Ceresa cycle is not trivial ([ELS24]), see Section 9.

5. Difference between $\mathbb{C}$ and $\mathbb{Q}$

Everything we said is true over arbitrary fields. So from now on assume we can also do everything over $\mathbb{Q}$.

The only thing that was very complex analytic is the construction of the Jacobian variety of a curve. Thankfully, even this was solved for arbitrary fields by André Weil in 1948 in his work on the Riemann hypothesis for curves over finite fields.

The difference between $\mathbb{C}$ and $\mathbb{Q}$ is substantial: over $\mathbb{C}$ we have Mumford's theorem. On the other hand, for X an algebraic variety over $\mathbb{Q}$, the chow groups are conjecturally (Beilinson–Bloch) finitely generated. Furthermore, the ranks of the chow groups are even computable (via certain L-functions).

6. PLAN FOR TALK

(1) start with homology (singular or simplicial) of topological spaces, it is defined by cycles mod boundaries. define algebraic variety. we are looking for an analogue.

(2) Examples: $\mathbb{P}^1$, Elliptic curve (of CH_0)

(3) If X over $\mathbb{C}$ then we also have usual homology and we actually have $\mathrm{CH}_i \to \mathrm{H}_i$ so this says that we got a good definition for CH_i.

(4) wild vs tame. wild: over $\mathbb{C}$ everything is complicated, Mumford's theorem and extensions say that this is a huge group. tame: over $\mathbb{Q}$ Beilinson–Bloch conjecture (an extension of Mordell–Weil) says that CH_i is finitely generated.

(5) give Jacobian as an example. give analytic construction over $\mathbb{C}$.

(6) If $X \subseteq \mathbb{P}^n$ quadric hypersurface, $\mathrm{CH}_0(X) = \mathbb{Z}$ but image of $\deg\colon \mathrm{CH}_0(X) \to \mathbb{Z}$ is either $\mathbb{Z}$ or $2\mathbb{Z}$. Springer's theorem, if $X \subseteq \mathbb{P}^3_{\mathbb{Q}}$ then $\mathrm{CH}_0(X) = \mathbb{Z}$ and image of deg might be $3\mathbb{Z}$. Theorem (D. Coray): If a cubic hypersdurface hnas a point of degree 3 then there exists effective cycle of degree 1 or 4 or 10. Voisin ('25) removed 10. Speculation for higher degree: image of deg is $d\mathbb{Z}$ where d is degree of X. obvious examples: intersection with hyperplane. arithmetic geometric variant: X_d the universal hypersurface. HKM's theorem. my problem.

7. Introduction to Algebraic Cycles

Let X be a smooth proper variety (integral scheme of finite type) over a field K. A natural way to study X is by studying its subvarieties and the interactions between them.

In the topological setting this leads to the theory of homotopy groups, singular homology, etc.

In the algebro-geometric setting the topology isn't rich enough for these notions to work as is. Instead, one develops the theory of algebraic cycles and define on them the equivalence relations known as *rational equivalence* and *algebraic equivalence*.

Given $i \in \mathbb{Z}_{\geq 0}$, an i-dimensional algebraic cycle (i-cycle, for short) is a formal sum of i-dimensional irreducible subvarieties. We denote by $Z_i(X)$ the set of all i-cycles in X (which is naturally an abelian group).

The degree map is a group morphism $\deg\colon Z_i(X) \to \mathbb{Z}$ defined by sending a formal sum to the sum of its coefficients. We say that an i-cycle $A \in Z_i(X)$ is effective if all of its coefficients are non-negative.

Remark. It is also sometimes useful to index by codimension instead of dimension. Every time an object is indexed by dimension (and if X is equidimensional), we use upper indices (instead of lower indices) to index by codimension. for example $Z^i(X) = Z_{\dim(X)-i}(X)$.

Example 13. $Z^1(X)$ is the group of Weil divisors.

Now we define two equivalence relations on $Z_k(X)$, motivated by their topological counterparts.

Definition 14 (Rational Equivalence, [Ful84] Proposition 1.6)**.** Rational equivalence is the transitive closure of the following equivalence: We say that $Y_1, Y_2 \in Z_k(X)$ are equivalent if there exists $W \subseteq X \times \mathbb{P}^1$ a $k+1$-dimensional subvariety whose projection $p\colon W \to \mathbb{P}^1$ is dominant and

$$Y_1 - Y_2 = p^{-1}(0) - p^{-1}(\infty)$$

Definition 15 (Algebraic Equivalence)**.** Replacing $\mathbb{P}^1$ with a general curve C (and $0, \infty$ with arbitrary points $p, q \in C$) in the definition of rational equivalence, we obtain a coarser equivalence relation called algebraic relation.

We define the i-dimensional Chow group of X, by $\mathrm{CH}_i(X) \coloneqq Z_i(X)/\sim_{\mathrm{rat}}$

Example 16. In codimension 1, rational equivalence is the same as linear equivalence from the classical theory of divisors and $\mathrm{CH}^1(X)$ is the divisor class group of X.

Remark. The definitions above can be extended to singular varieties.

7.1. Comparison with (co)homology.

If $K = \mathbb{C}$, we can view X as a real compact manifold and study $\mathrm{H}_\bullet(X)$, its singular/simplicial homology (or in general Borel-Moore Homology, if it isn't compact)

Claim 17. *([Ful84] Proposition 19.1.1) Rationally equivalent cycles are homologous.*

Corollary 18. *If X is a variety over $\mathbb{C}$, there is a map $\mathrm{CH}_i(X) \to \mathrm{H}^{BM}_{2i}(X, \mathbb{Z})$, defined by sending a subvariety Y to its fundamental class.*

Definition 19. The kernel of the above map, denoted by $\mathrm{CH}_{i,\mathrm{hom}}(X)$, is called the subgroup of homologically trivial cycles.

Definition 20. The Griffiths group of a variety X over $\mathbb{C}$ is defined by $\mathrm{Griff}_i(X) = \mathrm{CH}_{i,\mathrm{hom}}(X)/\sim_{\mathrm{alg}}$

7.2. **Examples of certain cycles.** Given a curve, there are certain constructions of cycles that were shown to have special properties and connections to other aspects of the given curve.

7.2.1. *Ceresa cycle.* Given a curve C, and a point P, we can embed C into it's Jacobian $\iota_P : C \to J$ by $\iota_P(Q) = [Q] - [P]$ (thinking of J as the picard variety). We then get another embedding by $-\iota_P$.

Definition 21. The Ceresa cycle of C is defined by

$$\kappa_P(C) \coloneqq [\iota_P(C)] - [-\iota_P(C)] \in \mathrm{CH}_1(J)$$

Theorem 22 (Ceresa, [Cer83])**.** *The Ceresa cycle of a "general" curve with genus ≥ 3 has infinite order in* $\mathrm{Griff}_1(J)$.

Claim 23. *Ceresa cycle of a hyperelliptic curve is trivial.*

Proving vanishing or non-vanishing of the Ceresa cycle has been a matter of interest recently, for example

Theorem 24 (Laga–Shnidman, [LS25])**.** *There is an injection* $\mathrm{H}^1(J)(-1) \hookrightarrow \mathrm{H}^3(J)$ *given by cupping with the principal polarization. Denote by* $H^3(J)_{\mathrm{prim}} \coloneqq H^3(J)/H^1(J)(-1)$ *its cokernel. If* $H^3(J)_{\mathrm{prim}}^{\mathrm{Aut}(C)} = 0$*, then* $\kappa_P(C)$ *is torsion.*

and on the other hand there is work to certify when the Ceresa cycle is not trivial ([ELS24]), see Section 9.

7.2.2. *Modified Diagonal cycle (AKA Gross-Schoen, Gross–Kudla–Schoen).* Let C be a (nice) curve and $b \in C$ We define diagonal cycles in $\mathrm{CH}^2(C \times C \times C)$ by

$$\begin{aligned} \Delta_{12,b} &= \{(x,x,b) \mid x \in C\} & \Delta_{13,b} &= \{(x,b,x) \mid x \in C\} & \Delta_{23,b} &= \{(b,x,x) \mid x \in C\} \\ \Delta_{1,b} &= \{(x,b,b) \mid x \in C\} & \Delta_{2,b} &= \{(b,x,b) \mid x \in C\} & \Delta_{3,b} &= \{(b,b,x) \mid x \in C\} \\ \Delta_{123} &= \{(x,x,x) \mid x \in C\} \end{aligned}$$

Definition 25 (Modified Diagonal cycle)**.** The b-pointed modified diagonal is defined by

$$\Delta_b \coloneqq \Delta_{123} - \Delta_{12,b} - \Delta_{13,b} - \Delta_{23,b} + \Delta_{1,b} + \Delta_{2,b} + \Delta_{3,b} \in \mathrm{CH}_2(C \times C \times C)$$

It is related to Beilinson-Bloch height pairing, triple product L-functions, etc.

As a matter of fact, the modified diagonal cycle is also related to the Ceresa cycle by the following claim:

Claim 26 (Zhang)**.** *The modified diagonal cycle is torsion in* $\mathrm{CH}_2(C^3)$ *if and only if the Ceresa cycle is torsion in* $\mathrm{CH}_1(J)$.

8. Rational equivalence of 0-cycles

Let F be a complete non-singular algebraic surface over $\mathbb{C}$. There are 3 equivalent notions of finite-dimensionality of $\mathrm{CH}_0(F)$:

(1) $\exists n$ such that $\forall A \in Z_0(F)$ with $\deg(A) \geq n$ there is some effective $A' \in Z_0(F)$ such that $A \sim_{rat} A'$.

(2) $\exists n$ such that the map $\mathrm{Sym}^n F \times \mathrm{Sym}^n F \to \mathrm{CH}_0(F)$ defined by $(A, B) \mapsto A - B$ is surjective.

(3) $\exists n$ such that for every m and $A \in \mathrm{Sym}^m F$ exists W a subvariety of $\mathrm{Sym}^m F$ of codimension at most n such that $A \in W$ and $A \sim_{rat} B$ for every $B \in W$.

The Italian school of algebraic geometry (or specifically Severi) assumed that the above conditions do hold, but as a matter of fact, Mumford proves in [Mum69] that for surfaces with positive genus, the above are necessarily false.

Mumford's proof follows from investigating differentials induced by certain maps.

Definition 27. Given smooth projective varieties X, Y and a morphism $\phi\colon Y \to \operatorname{Sym}^d X$, We define the trace map $\operatorname{Tr}_\phi\colon \mathrm{H}^0(X, \Omega^j) \to \mathrm{H}^0(Y, \Omega^j)$ by $\operatorname{Tr}_\phi(\omega) = \phi^*(\omega^{(d)})$, where $\omega^{(d)} \in \Omega^j_{X^d}$ is the form defined by $\omega^{(d)} = \sum_{i=1}^d p_i^*(\omega)$ where $p_i\colon X^d \to X$ is the projection on the i-th coordinate.

Note that $\omega^{(d)}$ is invariant under the S_d action on X^d by permuting coordinates. If it were a free action, we would've gotten that $\omega^{(d)}$ is a lift of a form from $X^d/S_d = \operatorname{Sym}^d X$. Strictly speaking, it defines a form on the non-singular locus of the action.

Theorem 28 ([Mum69], Roitman (reference TBD))**.** *Let X be a smooth projective variety, Y a smooth variety and $\phi\colon Y \to \operatorname{Sym}^d X$ be a morphism such that the class of $\phi(y) \in \mathrm{CH}_0(X)$ is independent of $y \in Y$ (meaning $\{\phi(y)\}_{y\in Y}$ are all rationally equivalent, when viewed as 0-cycles on X). Then $\operatorname{Tr}_\phi\colon H^0(X, \Omega^j) \to \mathrm{H}^0(Y, \Omega^j)$ vanishes for all $j > 0$.*

From this theorem we conclude that $\mathrm{CH}_0(F)$ is infinite-dimensional in the sense defined above.

9. Certifying nontriviality of ceresa classes of curves

[ELS24] describes an algorithm to check that the Ceresa cycle (or equivalently the modified diagonal cycle) is non-torsion in the Chow group. The main point is that they don't rely on symmetries of the curve, as opposed to previous works.

Let X be a nice curve over a number field K and $b \in X(K)$.

Definition 29 (b-pointed Z-shadow)**.** Let $Z \in \mathrm{CH}^1(X \times X)$ denote a correspondence on X, $\widetilde{Z} \coloneqq Z \times X$ and $\pi_3\colon X \times X \times X \to X$ the projection onto the third coordinate. We define the b-pointed Z-shadow

$$\operatorname{Sh}(Z, b) \coloneqq (\pi_3)_*(\Delta_b \times \widetilde{Z}) \in \mathrm{CH}^0(X)$$

Because Δ_b is homologically trivial, $\operatorname{Sh}(Z, b) \in \mathrm{CH}^0(X)_{\mathrm{hom}} = \operatorname{Pic}^0(X)$.

The shadow is a witness that Δ_b is non-trivial because

Lemma 30. *Let $m \in \mathbb{Z}$, if $m\operatorname{Sh}(Z, b) \neq 0$ in $\mathrm{CH}^0(X)$, then $m\Delta_b \neq 0$ in $\mathrm{CH}^2(X \times X \times X)$.*

and the point is that computations with the shadow are easier than with the modified diagonal.

Example 31. Specifically when taking $Z = Z_f$ to be the graph of a dominant map $f\colon X \to X$ when get interesting results, in particular:

(1) If $f\colon X \to X$ is the geometric Frobenius morphism, we get the b-pointed Frobenius shadow

$$\operatorname{Sh}(Z_f, b) = X(\mathbb{F}_q) - |X(\mathbb{F}_q)| b$$

Where $X(\mathbb{F}_q)$ is short for the divisor summing over all $\mathbb{F}_q$-points of X.

(2) If $f\colon X \to X$ is the identity, we get

$$\mathrm{Sh}(Z_f, b) = (2g-2)b - K_X$$

Thus, the above lemma recovers the known claim that Δ_b is nontrivial when $(2g-2) - K_X$ is nontrivial in $\mathrm{Pic}^0(X)$

The paper gives an algorithm which, when terminates, certifies that the modified diagonal cycle is non-torsion in $\mathrm{CH}^2(X \times X \times X)$.

Algorithm 32.

(1) Choose a nonempty finite set $\mathcal{T}$ of primes of K that are primes of good reduction for X.

(2) For every $\mathfrak{p} \in \mathcal{T}$, compute

$$N_{\mathfrak{p}} := \det(\mathrm{Frob}_{\mathfrak{p}} - I |\, \mathrm{H}^3_{\text{ét}}(X_{\overline{K}} \times X_{\overline{K}} \times X_{\overline{K}}, \mathbb{Z}_\ell(2))$$

and $M_{\mathfrak{p}}$ the order of the Frobenius shadow of the reduction of X at $\mathfrak{p}$.

(3) Define $\tilde{N} = \gcd_{\mathfrak{p}\in\mathcal{T}}(p^{\infty} N_{\mathfrak{p}})$ and $\tilde{M} = \mathrm{lcm}_{\mathfrak{p}\in\mathcal{T}}(p^{-\infty} M_{\mathfrak{p}})$. If $\tilde{M} \nmid \tilde{N}$, then the modified diagonal cycle of X has infinite order.

Remark. Under assumptions on the Sato-Tate group, the paper proves that if the image of Ceresa in étale cohomology is non-torsion then the algorithm will certify that in finite time.

10. A conjecture of Beilinson-Bloch, Bloch, and proving non-torsion in Griffiths group

A conjecture of Beilinson-Bloch states

Conjecture 33. *Let K be a number field and W a smooth, projective, geometrically irreducible variety over K, then*

$$\operatorname{rank} \mathrm{CH}^r(W)_{\mathrm{hom}} = \operatorname{ord}_{s=r} L(\mathrm{H}^{2r-1}(W), s)$$

Another conjecture of Bloch ([Blo84]), which "morally" follows from the previous conjecture is the following

Conjecture 34. *Suppose $\mathrm{H}^{2r-1}(W)$ motivically factors as $I \oplus M$ where M is of type $(2r-1,0)+(0,2r-1)$ and $\mathrm{h}^{2r-1,0}(I) = 0$, then*

$$\operatorname{rank} \mathrm{Griff}^r(W) = \operatorname{ord}_{s=r} L(M, s)$$

[BST97] gives empirical evidence to Theorem 34. The paper focuses on the case $r = 2$, $W = W_a := E_a^3$ where E_a ($a \in \mathbb{Q}$) is the elliptic curve over $\mathbb{Q}$ with affine model

$$E_a\colon \quad y^2 = (a^2-4)x^3 + 2a(a-2)x^2 + (a^2-4)x$$

For every $a \in \mathbb{Q}$, the non-singular genus 3 curve

$$C_a\colon \quad x^4 + y^4 + z^4 + a(x^2y^2 + y^2z^2 + z^2x^2) = 0$$

has 3 independent maps into E_a, which are used together to obtain an embedding of C_a into W_a. A 1-cycle $\Xi_a \in Z_1(W_a)$ is constructed from the previously mentioned embedding in a way akin to the Ceresa cycle.

To estimate the left hand side of the equation in Theorem 34, they give a lower bound on the rank by studying the class of Ξ_a in $\mathrm{Griff}_1(W_a)$. they prove two theorems (4.8 and 5.1), The first theorem establishes theoretical criteria for Ξ_a to have infinite order in $\mathrm{Griff}_1(W_a)$ and the second one reduces the criteria to calculation over $\mathbb{F}_{p^3}$, making it computable. Afterwards, they use an isogeny $E_{-3-a} \to E_a$ to push Ξ_{-3-a} to $\mathrm{Griff}_1(W_a)$ and prove in certain cases that it is linearly independent of Ξ_a, so that they get $\operatorname{rank} \mathrm{Griff}_1(W_a) \geq 2$ for some $a \in \mathbb{Q}$.

11. Further Directions/Questions

11.1. **Certifying nontriviality of Ceresa with respect to algebraic equivalence.** I will Combine ideas from [ELS24] and [BST97] to certify nontriviality of Ceresa/modified diagonal with respect to algebraic equivalence instead of rational equivalence.

Let C be a genus g curve over $\mathbb{Q}$ with Jacobian J.

Problem. Find sufficient conditions for the Ceresa cycle $\kappa(C) \in \mathrm{Griff}_1(J)$ to be non-torsion.

Suppose $\mathrm{H}^{2g-3}(J)$ motivically factors as $I \oplus M$ where M is of type $(2g-3, 0) + (0, 2g-3)$ and $\mathrm{h}^{2g-3,0}(I) = 0$. Theorem 34 says that if M is trivial, then $\mathrm{Griff}_1(J) = \mathrm{Griff}^{2g-1}(J)$ has rank 0 and so $\kappa(C) \in \mathrm{Griff}_1(J)$ is torsion if and only if $\kappa_P(C) \in \mathrm{CH}_1(J)$ is torsion. This means that the algorithm in [ELS24] already solves the above problem, conditional on Theorem 34.

Problem. If J is hodge generic, is it true that Ceresa is torsion in Griffiths if and only if it is torsion in Chow?

Because of the above discussion, we want to restrict to C with M non-trivial.

We can start by investigating specific families of genus 3 curves over $\mathbb{Q}$. They are classified by their Jacobian:

(1) Jacobian with real multiplication.
(2) Jacobian with action by an imaginary quadratic field.
(3) Jacobian with a map to an elliptic curve (i.e split)

We will treat a specific case of the third family above.

Example 35. The curves C_a in [BST97] are exactly the 1-dimensional family of curves with automorphism group S_4 [LMFDB, 3.24-12.0.2-2-2-3.1]. This family is contained in a 2-dimensional family of curves with automorphism group S_3 [LMFDB, 3.6-1.0.2-2-2-2-3.1] with parametrization

$$C_{a_1,a_2}: \quad x^4 + y^4 + z^4 + a_1(x^2y^2 + x^2z^2 + y^2z^2) + a_2(x^2yz + y^2xz + z^2xy) = 0$$

(Note that $C_a = C_{a,0}$)

Problem. Develop a method like in [ELS24] for the S_3-family in the example above. This will be a generalization of [BST97].

Problem. Identify for which curves in the family M is non-trivial.

Problem. Work out if the method of shadows as in [ELS24] works for algebraic equivalence.

Question. Is it true that Ceresa cycle and the modified diagonal cycle are torsion/non-torsion together in the Griffiths groups (similarly to Chow)?

11.2. **Low Degree subvarieties of universal pointed hypersurfaces.** Let $X_{n,d}$ be the universal complex degree d hypersurface of dimension n, i.e the hypersurface defined by $\sum a_I x^I = 0$ in variables $x_0, \dots, x_{n+1}$ over $\mathbb{C}(a_I)$, where I runs over multi-indices of degree d.

Standard computations show that $X_{n,d}$ is of index d, meaning every cycle on $X_{n,d}$ has degree divisible by d. One way of obtaining a cycle on $X_{n,d}$ of degree kd is by intersecting $X_{n,d}$ with a general degree k subvariety. Recently [HKM25] proved that for $d \gg n, k$ every degree kd effective cycle on $X_{n,d}$ is contained in a degree k subvariety none of whose components lie on $X_{n,d}$.

In short, the method presented in [HKM25] starts with reducing the general case (of cycles of arbitrary dimension) to the case of 0-cycle. The main ingredients for the case of 0-cycles, is the following claim about Cayley-Bacharach condition (and a claim relating CB to covering by curves)

Definition 36. We say that that a collection of points in projective space satisfies $CB(r)$ for some $r \in \mathbb{N}$ if any degree r hypersurface passing through all but one of the points passes through all of them.

Claim 37 ([HKM25], Proposition 1.5)**.** *Let $X \subseteq \mathbb{P}^{n+1}_{\mathbb{C}}$ be a very general hypersurface of degree $d \geq 2n+3$. If an effective 0-cycle $p_1 + \ldots + p_{kd} \in Z_0(X)$ of degree kd is rationally equivalent to $k \cdot c_1(\mathcal{O}_X(1))^n$, then a subset of $\{p_1, \dots, p_{kd}\}$ satisfies $CB(d-2n-2)$*

The proof of the above claim uses the following theorem of Riedl–Yang.

Theorem 38 ([RY21], Theorem 2.4)**.** *Fix a positive integer n_0 and consider for all $1 \leq m \leq n_0$ loci $Z_{m,d} \subseteq \mathcal{X}_{m,d}$ which are countable unions of locally closed subvarieties and satisfy:*

(1) *If $(p, X) \in Z_{m,d}$ is a parametrized hyperplane section of $(p', X') \in \mathcal{X}_{m+1,d}$ then $(p', X') \in Z_{m+1,d}$.*

(2) *$Z_{n_0-1,d}$ has codimension at least one in $\mathcal{X}_{n_0-1,d}$*

Then the codimension of $Z_{n_0-c,d}$ in $\mathcal{X}_{n_0-c,d}$ is at least c.

I will research what happens when we replace $X_{n,d}$ with the universal degree d pointed hypersurface of dimension n.

Definition 39. Denote by $X^*_{n,d}$ the universal complex degree d pointed hypersurface of dimension n. Concretely, it is the generic fiber of the universal family of complex degree d pointed hypersurface of dimension n $\mathcal{X}^*_{n,d} \to B^*_{n,d}$ where and

$$B^*_{n,d} \coloneqq \mathbb{P}\,\mathrm{H}^0(\mathbb{P}^{n+1}_{\mathbb{C}}, \mathcal{O}(d)) \times \mathbb{P}^{n+1}_{\mathbb{C}},$$
$$\mathcal{X}^*_{n,d} \coloneqq \{(x,(f,y)) \mid f(x)=0,\ f(y)=0\} \subseteq \mathbb{P}^{n+1}_{\mathbb{C}} \times B^*_{n,d}$$

Problem. It is no longer true that $X^*_{n,d}$ has index d, for example because by construction it has a degree 1 point. Still, some degrees do not appear as intersections of curves, can they be classified? Try to eliminate certain degrees.

Problem. Is there a variant of Theorem 38 for the universal pointed hypersurface?

Problem. Compute $\mathrm{CH}_0(\mathcal{X}^*_{n,d})$.

Problem. Start with the specific case of $n=1$. In [HKM25] Lemma 1.4 they show a short proof for their result for $n=1$, using projective normality. Could something similar be done for the universal pointed curve $X^*_{1,d}$?

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